Generative vs discriminative: generative model learns $p(x, y)$; classification recoverable via Bayes $P(y|x) = p(x, y)/p(x)$. Discriminative learns only $P(y|x)$; cannot generate x samples.

Memorisation vs generalisation: exact memorisation achieves zero reconstruction error but cannot produce novel samples outside the training set. A generative model must assign high probability to plausible unseen inputs.

Generative paradox: simultaneous tensions – learn patterns without memorising; coherent without repetitive; diverse without losing structure; realistic without overfitting; novel without random. VAE resolves via KL regularisation; GAN via adversarial loss; diffusion via noise-injection/reconstruction.

Generative AI vs Generative ML: broad = any system producing novel artefacts (rule-based, evolutionary, CSP) – no data required. Narrow (ML) = learns $p(x)$ from examples and samples from it. AARON is generative AI but not generative ML.

Applications: anomaly detection (likelihood scoring); missing data imputation (sample plausible completions); data augmentation (balance class distributions).

AARON: symbolic rule-based painter; all aesthetic knowledge hand-coded by Cohen. Novelty comes from rule combinations, so outputs cannot exceed the designer's encoded knowledge. Creativity is bounded.

L-systems: parallel rewriting – all symbols rewritten simultaneously at each step. Models biological growth where every cell divides in lockstep, producing self-similar fractal structures (plants, ferns). Sequential rewriting would break self-similarity since some branches would always be one derivation ahead.

Illiac Suite: generate-then-test loop – stochastic process proposes random notes; rule-based screener applies counterpoint constraints to accept or reject. Randomness = exploration; rules = musical validity.

Evolutionary art (Sims): fitness function + genetic operators; no behavioural knowledge hand-coded. Produces morphologies the designer never imagined – genuine surprise. Contrast with AARON: evolutionary systems can exceed designer knowledge through search.

k-gram models: V^{k-1} possible contexts grow exponentially; most transitions unseen in training data. RNNs use a compressed hidden state instead of enumerating all contexts, removing the sparsity bottleneck.

Denoising AE: corrupts input before encoding; forces model to learn the clean-data manifold rather than the identity. Implicitly estimates $\nabla_x \log p(x)$ (same signal diffusion models use explicitly).

Sparse AE: overcomplete latent ($\dim z \geq \dim x$); L1/KL/top-k penalty limits active units. Learns richer vocabulary than a bottleneck.

Contractive AE: penalises Frobenius norm of encoder Jacobian $\|J_f(x)\|_F^2$; small norm = output insensitive to input perturbations; prevents copy by making representation locally invariant.

U-Net: encoder–decoder with skip connections that pass high-res spatial features directly to decoder; pooling discards fine detail, skip connections restore it. Used as the denoising network in diffusion models (t injected via sinusoidal embeddings, conditioning c via cross-attention).

ELBO: $\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z))$. Term 1 reconstruction; term 2 pulls encoder posterior toward $\mathcal{N}(0, I)$.

Reparameterisation: $z = \mu_\phi(x) + \sigma_\phi(x) \odot \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$. Moves stochasticity to ε (no parameters), so gradients flow through μ_ϕ, σ_ϕ.

Why VAE latent space is organised: KL term forces every encoder output toward $\mathcal{N}(0, I)$, so all image codes cluster near the origin; sampling from $\mathcal{N}(0, I)$ always lands in a region the decoder was trained on. GAN has no encoder and no reconstruction requirement, so the generator's mapping is arbitrary and no region of $\mathcal{Z}$ is guaranteed to decode meaningfully.

Posterior collapse: KL dominates; encoder ignores input, decoder outputs training-data average. Fix: KL annealing (weight starts at 0, increases over early epochs).

β-VAE: $\beta > 1$ on KL term encourages disentanglement (independent latent dims) at the cost of reconstruction quality.

VQ-VAE: discrete codebook; nearest-neighbour lookup replaces continuous latent; eliminates posterior collapse; requires learned prior (PixelCNN/Transformer) for generation. Gradient problem: $\arg\min$ has zero gradient, so straight-through estimator is used – gradients from reconstruction loss pass from z_q directly to z_e as if quantisation were identity.

VAE blur (MSE): MSE reconstruction loss penalises all pixel deviations equally, so the model minimises expected error by outputting the average of all plausible reconstructions – spreading probability mass over valid configurations produces blurry outputs. Perceptual loss (VGG features) targets structure over pixel values; adversarial loss produces sharp outputs but may hallucinate incorrect details.

Classical AE sampling problem: no defined distribution over latent space; random $z \sim \mathcal{N}(0, I)$ hits gaps, producing garbage. VAE's KL regularisation is precisely what prevents this.

GAN minimax: $\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$. Optimal $D^*(x) = p_{\text{data}}/(p_{\text{data}} + p_g)$; substituting D^* gives JS divergence. At equilibrium $D = 0.5$ everywhere, $p_g = p_{\text{data}}$.

Non-saturating loss: when $D(G(z)) \approx 0$, $\log(1 - D(G(z))) \approx 0$ (vanishing gradient). Fix: G maximises $\log D(G(z))$ instead; same fixed point, much better early gradient.

JS vs KL: KL asymmetric; JS symmetric via $M = \frac{1}{2}(P + Q)$, bounded by $\log 2$. JS emerges naturally from the GAN loss (not a design choice). JS saturates to $\log 2$ when distributions have disjoint support – no gradient signal.

Wasserstein-1 (Earth Mover's): cost of moving mass from p_g to p_{data}; gives meaningful gradient even when distributions don't overlap. Requires critic to be 1-Lipschitz: $|f(x_1) - f(x_2)| \leq \|x_1 - x_2\|$.

WGAN: critic outputs unbounded score (not probability); score difference $\approx W_1$ via Kantorovich–Rubinstein dual. Weight clipping enforces Lipschitz crudely (distorts capacity).

WGAN-GP: penalises $\|\nabla f(\hat{x})\|_2 \neq 1$ on interpolated points $\hat{x} = \epsilon x_{\text{real}} + (1 - \epsilon) x_{\text{fake}}$; smoother, principled constraint that does not distort network capacity.

Mode collapse: G maps many z to a small output set that reliably fools D; D adapts, G jumps to another mode; cycles without convergence. D cannot directly penalise missing modes since it sees individual samples, not the full output distribution. Detection: generate large batch, check low variety; apply classifier and check few classes appear. Mitigations: (1) WGAN; (2) minibatch discrimination (D sees group statistics, detects low diversity); (3) feature matching (G matches D's intermediate feature statistics on real data).

DCGAN: strided convolutions instead of pooling (learnable downsampling); BatchNorm everywhere except G output and D input; no FC layers (respects spatial structure); LeakyReLU ($\alpha \approx 0.2$) in D (prevents dead neurons), ReLU in G.

CycleGAN: unpaired images; 4 networks (G_{AB}, G_{BA}, D_A, D_B); cycle consistency $\|G_{BA}(G_{AB}(x)) - x\|_1$ prevents mode collapse by penalising information loss. **Pix2Pix:** paired images; 2 networks;

L1 + adversarial; PatchGAN discriminator.

StyleGAN: mapping network $\mathcal{Z} \to \mathcal{W}$ (8-layer MLP). $\mathcal{Z}$ Gaussian constraint forces feature entanglement; $\mathcal{W}$ is free from the prior and can be more disentangled. Style injected at multiple resolutions via AdaIN. Editing directions in $\mathcal{W}$ produce cleaner semantic changes.

BigGAN: class conditioning + truncation trick (sample z from truncated Gaussian for fidelity/diversity tradeoff).

DDPM forward: $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$. Schedule $\{\beta_t\}$ increases ($0.0001 \to 0.02$): small early β_t destroys detail first, large late β_t destroys coarse structure. One-shot formula: any (x_t, t) pair sampled directly from x_0 without stepping through the chain.

DDPM training: U-Net predicts ε from (x_t, t); loss $\mathbb{E}\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$. Predicting ε preferred over μ: zero-mean target regardless of t, consistent scale across 1000 steps.

DDPM reverse: predict ε_θ, compute μ_θ, sample $x_{t-1} \sim \mathcal{N}(\mu_\theta, \sigma_t^2 I)$; fresh Gaussian noise added at each step (except $t = 1$); variance fixed from schedule, not predicted.

DDIM: deterministic ODE sampler on same DDPM-trained model. Same $x_T \to$ same image (invertible). Smooth trajectory allows larger steps: 50–100 instead of 1000. DDIM inversion derived for $w = 1$; at $w > 1$ (standard CFG), inversion and sampling trajectories diverge – reconstruction at non-trivial guidance scale fails without additional correction (e.g., null-text optimisation).

Classifier guidance (CG): separate classifier trained on noisy images; at each denoising step: $\hat{\varepsilon} = \varepsilon_\theta(x_t, t) - s\sqrt{1-\bar{\alpha}_t}\, \nabla_{x_t} \log p(y|x_t)$. Scale $s = 0$ = unconditional. Requires extra network; does not scale to text.

CFG: condition dropout ($\sim 10\%$) during training; single model learns $\varepsilon_\theta(x_t, t, c)$ and $\varepsilon_\theta(x_t, t, \emptyset)$. Sampling: $\hat{\varepsilon} = \varepsilon_\theta(\emptyset) + w(\varepsilon_\theta(c) - \varepsilon_\theta(\emptyset))$. $w > 1$ extrapolates toward conditional direction; higher w: sharper but less diverse, can oversaturate. c injected via cross-attention; t via sinusoidal embeddings.

Imagen: cascade of diffusion models – base model at low resolution, followed by super-resolution diffusion models. Demonstrates that large frozen text encoders (T5-XXL) provide stronger text conditioning than jointly trained encoders.

VAE vs GAN vs DDPM: VAE – good stability, blurry outputs, good mode coverage, fast (1 pass), approx likelihood (ELBO). GAN – poor stability (mode collapse, vanishing gradients), sharp outputs, mode collapse risk, fast (1 pass), no likelihood. DDPM – excellent stability (MSE regression), high quality + diverse, excellent mode coverage, slow (1000 steps; 50 with DDIM), approx likelihood.

LDM (Latent Diffusion): frozen VAE encodes $512\times512\times3 \to 64\times64\times4$ ($\approx 48\times$ compression); DDPM runs in latent; frozen VAE decoder reconstructs. Self-attention scales quadratically with spatial dim: $8\times$ spatial reduction $\to 64\times$ fewer tokens.

FID: fits InceptionV3 features of real and generated sets as Gaussians; Fréchet distance between them. Assumes Gaussianity (breaks for multimodal distributions, inflates covariance). Cannot separate quality from diversity. Requires >10k samples; sensitive to preprocessing choices (resize algorithm, normalisation).

Precision/Recall: Precision = fraction of generated samples on real manifold (fidelity). Recall = fraction of real manifold covered by generated samples (diversity). FID can miss mode collapse when quality is high but diversity is low; Precision/Recall separates the two.

KID: kernel-based, no Gaussian assumption, unbiased, reliable at small sample counts.

CLIPScore: cosine similarity in CLIP shared space between image and text. Misses: realism, spatial reasoning, fine-grained composition. Pair with FID.

IS: no real data reference; quality (classifier confidence) + diversity (marginal entropy). Gameable by ImageNet training; mixed signal with no distributional comparison.

Protocol issues: resize algorithm, feature extractor version, sample count, and sampling hyperparameters all shift FID; cross-paper comparison unreliable without identical protocol.

Lerp vs Slerp: linear interpolation (lerp) in high-D Gaussian space passes through the low-density interior (midpoint norm $\approx \|z\|/\sqrt{2}$, well below the $\sqrt{d}$ shell where mass concentrates) – produces incoherent outputs. Slerp follows the great-circle arc on the hypersphere, keeping norm constant and staying in high-density regions throughout the path.

GAN inversion trade-off: optimise z^* minimising $\|G(z^*) - x\|$. Aggressive reconstruction pushes z^* into $\mathcal{W}^+$ (per-layer space with more capacity), outside the region the model was trained on. Semantic editing directions calibrated for $\mathcal{W}$ become unreliable there.

Prompt-to-Prompt: freeze cross-attention maps from source generation; swap text tokens. Spatial layout (which positions attend to which tokens) is preserved; rendered content changes.

DDIM inversion + CFG: inversion formula assumes $w = 1$; at $w > 1$, inversion trajectory diverges from sampling trajectory, accumulated error prevents faithful reconstruction.

ControlNet: frozen U-Net + trainable parallel encoder clone; zero-init convolutions connect clone outputs to frozen decoder. At step 0, zero convolutions zero out clone contribution – model starts as pure frozen base, base quality cannot degrade during early training.

Textual Inversion: learns single token embedding S^* (∼KB); all model weights frozen. Learns the correct embedding in the current model's representation space. Moderate fidelity; high portability.

DreamBooth: fine-tunes full U-Net (or LoRA delta) on 3–5 images; binds subject to a rare identifier token. Class-preservation loss simultaneously trains on synthetic images of the general class to prevent catastrophic forgetting.

InterFaceGAN: SVM on labelled latent codes; decision boundary normal = editing direction (named, interpretable, requires labels).

GANSpace: PCA on latent codes; principal axes = dominant visual variations (label-free, less interpretable).

Autoregressive factorisation: $P(x) = \prod_t P(x_t|x_{<t})$; exact, no independence assumption; generation is sequential – cannot parallelise ($O(T)$ forward passes).

Temperature: divides logits by τ before softmax. $\tau < 1$: sharper (deterministic); $\tau > 1$: flatter (diverse).

Top-k: restrict to k most probable tokens; fixed cutoff ignores distribution shape. **Nucleus (top-p):** keep smallest token set summing to probability p; adapts to model confidence. Combined with temperature in practice (temperature first, nucleus cuts tail).

Beam search: heuristic best-first search over top-k partial sequences; not exhaustive; tends toward repetitive output.

Hallucination: factual (invents facts contradicting reality); faithfulness (contradicts provided context). Root cause: autoregressive training rewards next-token prediction, not factual accuracy – truth is not in the training signal.

Chain-of-thought: intermediate tokens serve as working memory; each reasoning step conditions subsequent ones; activates step-by-step reasoning pattern from pretraining data.

Zero-shot / few-shot / ICL: instruction only vs. 2–10 in-context examples vs. umbrella term for any context-driven adaptation without weight updates.

BPE: iteratively merge most-frequent adjacent character pair until vocabulary reaches target size; variable-length subwords; avoids OOV; shorter sequences than character-level.

Rectified flow: interpolant $x_t = (1-t)x_0 + tx_1$, $x_1 \sim \mathcal{N}(0, I)$. Target velocity $u_t = x_1 - x_0$ (constant per pair). Loss: $\mathbb{E}\|v_\theta(x_t, t) - (x_1 - x_0)\|^2$. Straight-line paths allow larger ODE integration steps: DDPM's curved paths accumulate error per step, so many small steps are needed; straight-line Euler integration has near-zero approximation error at large steps. DDIM is a different sampler over a DDPM-trained model; rectified flow is a different training objective entirely.

Flow matching: general framework; rectified flow = linear interpolant; DDPM = variance schedule as probability path. Flow matching is strictly more general.

Consistency models: $f_\theta(x_t, t) = x_0$ for all t on a trajectory; one-step generation from x_T. Self-consistency loss: $\|f_\theta(x_t, t) - f_{\theta^-}(x_{t'}, t')\|$ for adjacent t, t' on same trajectory (both should output same x_0). **CT:** compute $x_{t'}$ from forward formula (train from scratch). **CD:** compute $x_{t'}$ via one reverse step of pre-trained teacher; better targets $\rightarrow$ higher quality than CT. Multi-step refinement: x_0 estimate $\rightarrow$ re-noise to $x_{t'}$ $\rightarrow$ denoise again; trades speed for quality.

LoRA: freeze base model; add trainable $\Delta W = BA$ alongside each weight matrix (B tall-thin, A wide-thin; rank $\ll \min(d_{\text{in}}, d_{\text{out}})$). Assumes task-specific change lives in a low-dimensional subspace. ∼MB vs GB for full fine-tune; swappable per task.

DiT: transformer on latent patches. adaLN-Zero: MLP maps $(t, \text{class}) \rightarrow (\gamma, \beta)$ for LayerNorm modulation; residual gate α initialised to 0 (blocks start as identity, stable gradients from step 1). **MMDiT (SD3/FLUX):** two token streams (image patches + text); full joint attention in every layer so every image token attends to every text token and vice versa; deeper text influence than cross-attention alone.